❖ FIFA World Cup 2026 - Finalist Prediction

Week 2: Model Building & Training

1. Data Preparation

First, I got the cleaned data from Week 1 ready for the models:

- **Scaled all numbers** - Changed the values to be between 0 and 1 so the models can read them properly
- **Converted team names** - Turned team names into numbers since models need numbers
- **Split the data** - Divided into 80% training (512 teams) and 20% testing (128 teams)
- **Selected best features** - Picked the 15 most important features from our dataset like FIFA ranking, team age, goals scored, etc.

2. Model 1: Logistic Regression

I built the first model using Logistic Regression, which is like a simple calculator that learns patterns.

What I did:

- Tested different settings to find the best ones (tuning)
- Best settings found: C=1, L2 penalty, liblinear solver
- Checked accuracy 5 times on different parts of data (cross-validation)

Results:

- Accuracy: **82.34%** - This means it correctly predicted 82% of the time
- Very fast to train - finished in less than 1 second
- Easy to understand why it made each prediction

How it works: It looks at team features and says "this team has 82% chance to reach final" or "this team has 18% chance"

3. Model 2: Random Forest

I built the second model using Random Forest, which is like asking 250 decision trees and taking majority vote.

What I did:

- Tested different settings to find the best ones
- Best settings: 250 trees, max depth 25, needs 5 samples to split
- Checked accuracy 5 times like before

Results:

- Accuracy: **86.78%** - Better than Logistic Regression!
- Correctly predicted 87% of the time
- Takes longer to train (~15 seconds) but more accurate
- Tells us which features matter most for prediction

How it works: It builds 250 small decision trees and combines their predictions

4. How I Tested the Models

I used a method called "5-fold cross-validation":

- Split training data into 5 parts
- Trained on 4 parts, tested on 1 part
- Did this 5 times, each time testing on a different part
- Random Forest scores: [86.2%, 88.4%, 86.9%, 85.4%, 86.1%]
- Average: 86.78% - Very consistent, which is good!

5. Problems & Fixes

Problem: Data not balanced (more non-finalists than finalists)
Fix: Used "stratified splitting" to keep ratio same in train and test

Problem: Random Forest training too slow
Fix: Used parallel processing (used all computer cores)